Anton Paar Kaomi for Nova

version 1.0

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Report date: 08/13/2026 **Operator:** 78442
File Name: CNT.qcuPhysIso

Analysis Data

Sample

ID	CNT	**Weight**	0.1322 g		
Description	CNT				

Analysis

Data ID	{3d916f1c-fe01-40c0-b3f1-2ef8525a0c27}				
Analysis Profile	UJ N2 full isotherm(2)				
Operator	Solomon	**Date**	08/12/2026	**Duration**	811.90 min
Instrument	St 3 on NOVA 800 [s/n:1050040039]			**Firmware**	3.01
Comments:	""				
Ambient Temp.	0.00 °C	**Void Volume Mode**	He Measure	**Cold Zone**	25.2762 cm³
Warm Zone	3.08828 cm³	**Cell Type**	9 mm with filler rod		
Thermal Delay	180 sec	**p_0 Mode**	Monitor		

Adsorbate

Name	Nitrogen	**Molecular Weight**	28.0134 g/mol	**Cross Sectional Area**	16.2 Å²/molecule
Non-Ideality	6.58e-05 1/Torr	**Bath Temperature**	77.35 K		

Degas information

Type	Vacuum Degassing
Operator	Solomon
Description	CNT
Degassed	at 0 °C for 0 hours

Data Reduction Parameters

Data Reduction Parameters

Thermal Transpiration	no
Thickness Method	deBoer

Adsorbate Model

Name	Nitrogen	**Molecular Weight**	28.0134 g/mol	**Cross Sectional Area**	16.2 Å²/molecule
Bath Temperature	77.35 K				

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Isotherm-Relative

Relative Pressure, p/p_o	Volume Adsorbed @STP cm^3/g
0.00492589	7.73525
0.0162868	9.65248
0.0268984	10.5027
0.0522114	11.7466
0.0740909	12.5107
0.122913	13.9046
0.172949	15.1550
0.222860	16.3623
0.273014	17.6093
0.299631	18.3079
0.324919	18.9358
0.371370	20.1251
0.422090	21.5745
0.471695	23.1022
0.521036	24.6783
0.570550	26.5047
0.619749	29.0886
0.667566	31.5188
0.715496	34.5763
0.771714	40.3875
0.820299	47.5771
0.868478	58.9621
0.915860	81.6052
0.967899	157.120
0.991582	266.546
0.979704	264.692
0.937849	170.217
0.884311	101.530
0.833688	70.8177
0.783057	55.0821
0.736026	46.2775
0.681087	39.4288
0.625708	34.7349
0.581967	31.9853
0.533391	29.4159
0.482056	27.1135
0.431354	25.1726
0.381165	23.3992
0.331455	21.5207
0.280400	19.9887
0.229794	18.6130
0.179697	17.2692
0.123025	15.7050
0.0747224	14.2045
0.0214278	11.5964

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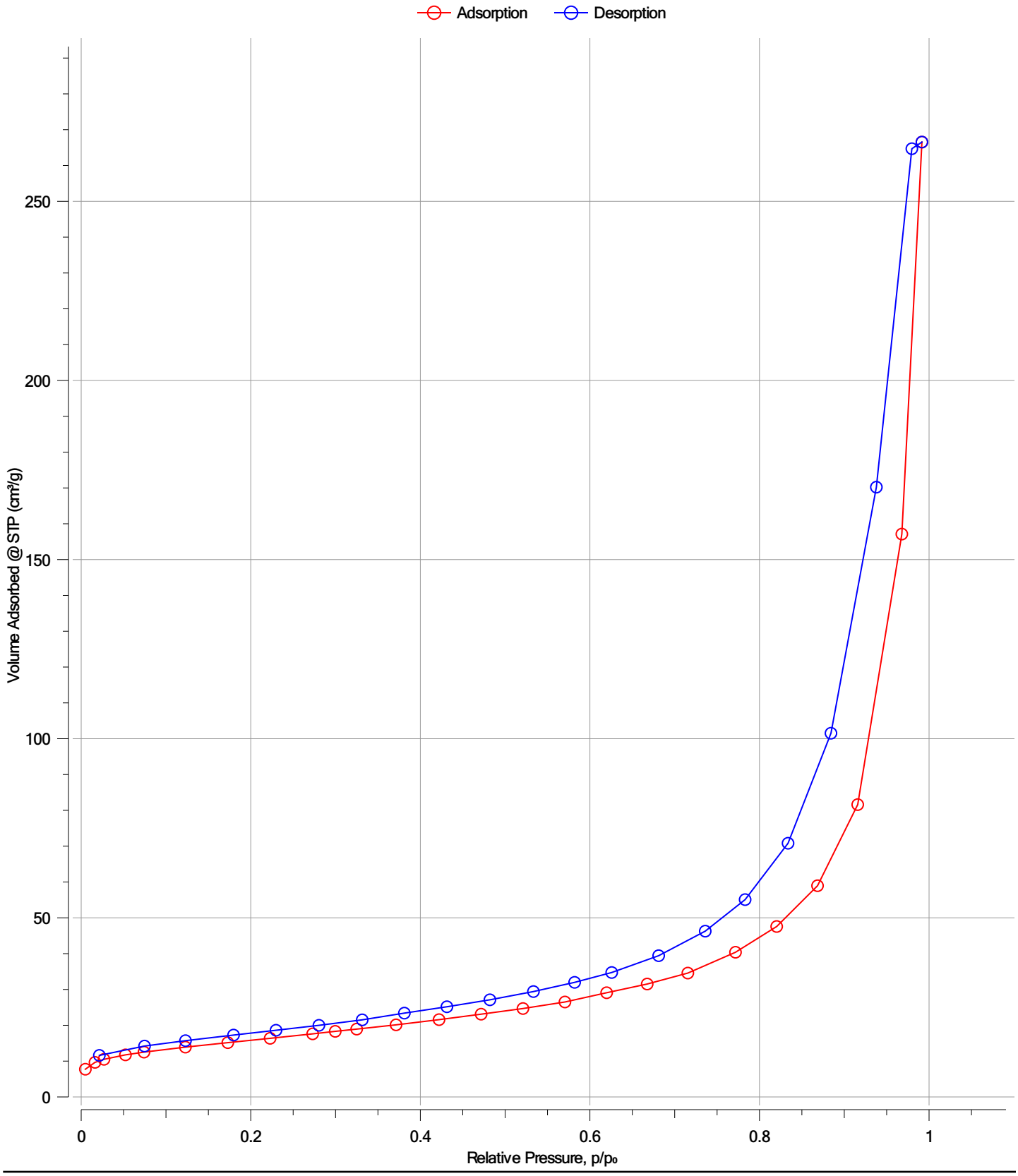
Isotherm-Linear



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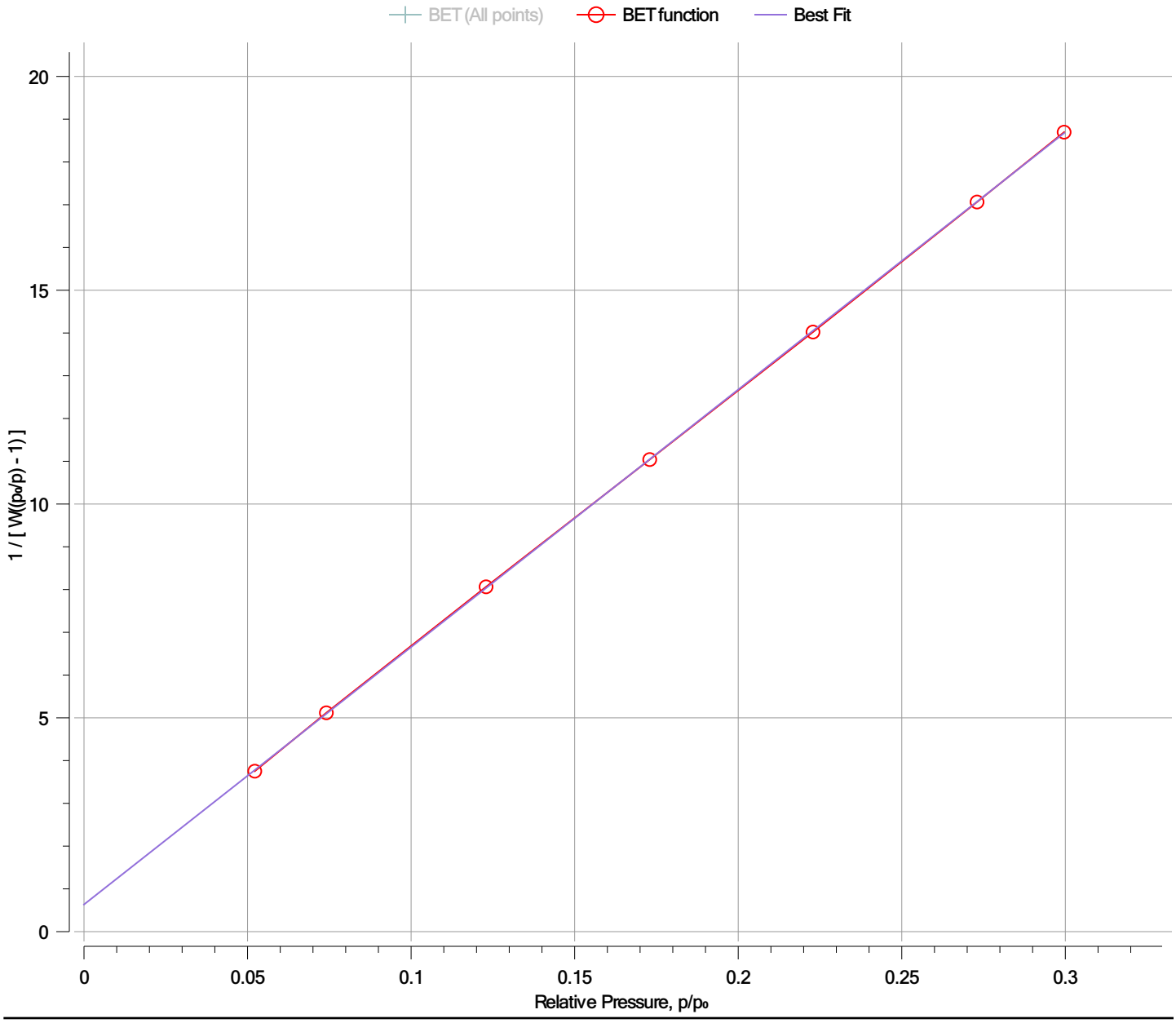
Multipoint BET Summary/Results

Isotherm Branch	Adsorption	**Slope**	60.207	**Intercept**	0.635023
Correlation coeff., r	0.999991	**C constant**	95.8108	**Surface area**	57.239 m²/g

BET-Multipoint BET

Relative Pressure, p/p_o	Volume Adsorbed @STP cm^3/g	$1 / [W((p_o/p) - 1)]$
0.0522114	11.7466	3.7523
0.0740909	12.5107	5.1176
0.122913	13.9046	8.0639
0.172949	15.1550	11.0402
0.222860	16.3623	14.0229
0.273014	17.6093	17.0634
0.299631	18.3079	18.6970

BET-Multipoint BET



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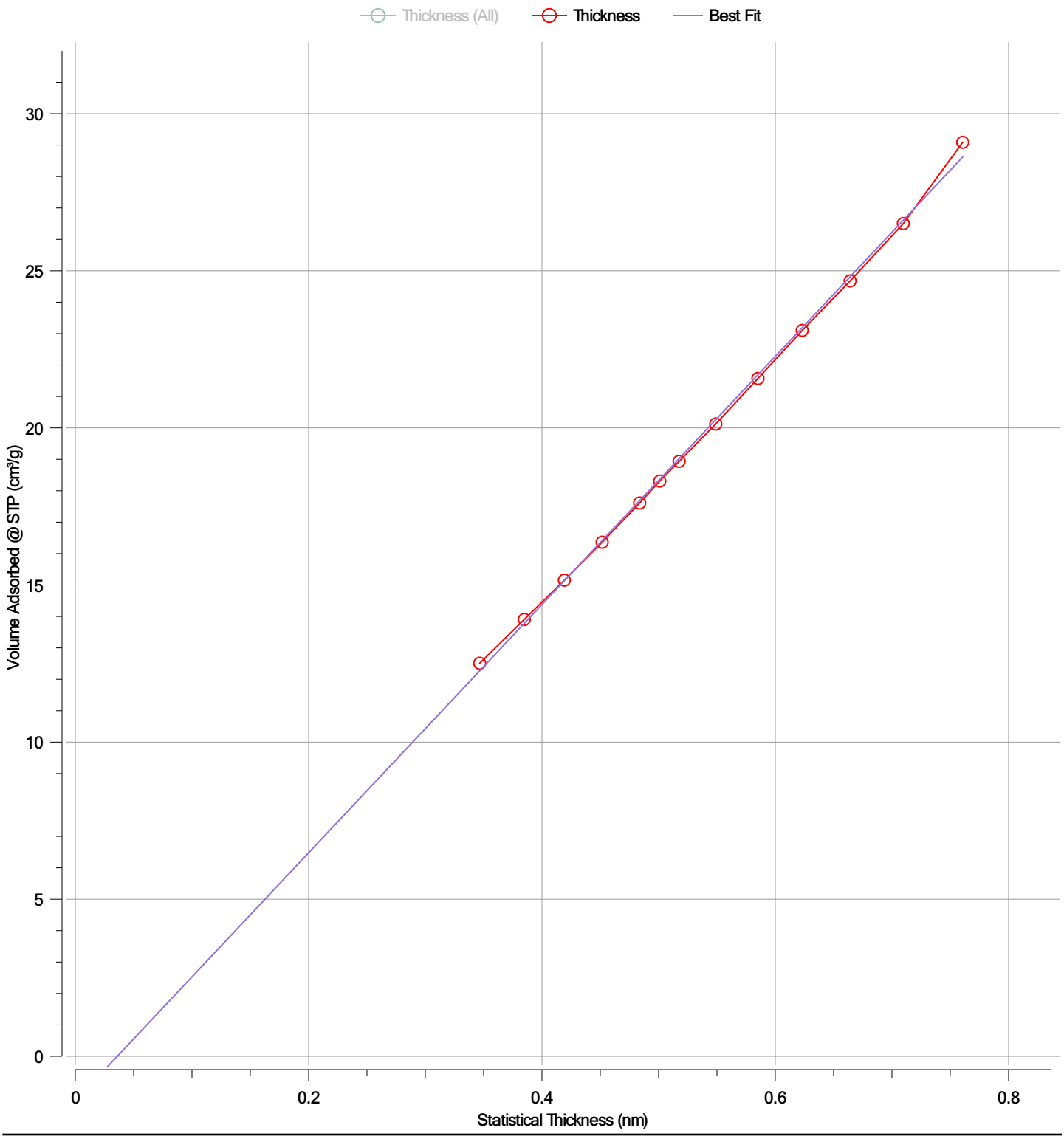
t-plot Method Summary/Results

Isotherm Branch	Adsorption	**Slope**	39.4938 cm³/nm/g	**Intercept**	-1.41998 cm³/g
Correlation coefficient, r	0.999364	**Thickness method**	deBoer	**Micropore volume**	0 cm³/g
Micropore area	0.000 m²/g	**External surface area**	57.239 m²/g		

t-plot Method



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t-plot Method

Relative Pressure, p/p_o	Volume Adsorbed @STP cm^3/g	Statistical Thickness nm
0.0740909	12.5107	0.3466
0.122913	13.9046	0.3849
0.172949	15.1550	0.4192
0.222860	16.3623	0.4516
0.273014	17.6093	0.4838
0.299631	18.3079	0.5010
0.324919	18.9358	0.5176
0.371370	20.1251	0.5490
0.422090	21.5745	0.5851
0.471695	23.1022	0.6231
0.521036	24.6783	0.6642
0.570550	26.5047	0.7098
0.619749	29.0886	0.7607